

GTIIT Spring 2025 Calculus 1M

Workshop 4 - Friday

March 31, 2025

Try to solve the following exercises. Feel free to chat about them with your classmates, but make sure to write your own solutions. If you get stuck, just move on to the next one and come back later.

If you need any help at all —whether it's a tiny hint or a big clue— don't hesitate to ask me!

Important note: If you find any errors or typos, please let me know so I can correct them.

1. Use mathematical induction (p. 72 Stewart's book) to show that if $f(x) = xe^x$, then $f^{(n)}(x) = (x+n)e^x$.
2. Suppose that:

$$f(1) = 2, \quad f'(1) = 3, \quad f(2) = 3, \quad f'(2) = 2, \quad g(1) = 3, \quad g'(1) = 1, \quad g(2) = 1, \quad g'(2) = 4,$$

- (a) If $S(x) = f(x) + g(x)$, find $S'(1)$.
- (b) If $P(x) = f(x)g(x)$, find $P'(2)$.
- (c) If $Q(x) = \frac{f(x)}{g(x)}$, find $Q'(1)$.
- (d) If $C(x) = fg(x)$, find $C'(2)$.

3. Calculate y' :

- (a) $y = \tan\left(\frac{t}{t^2+1}\right)$
- (b) $xe^y = y \sin(x)$
- (c) $y = e^{\cos x} + \cos(e^x)$
- (d) $y = \ln \sin(x) - \frac{1}{2} \sin(x^2)$

4. (a) Find an equation of the tangent line to the curve $y = e^x$ that is parallel to the line $1 = x - 4y$.
- (b) Find an equation of the tangent line to the curve $y = e^x$ that passes through the origin.
- (c) Find f' in terms of g' where $f(x) = \ln |g(x)|$.
- (d) Find h' in terms of f' and g' where $h(x) = \frac{f(x)g(x)}{f(x)+g(x)}$.